

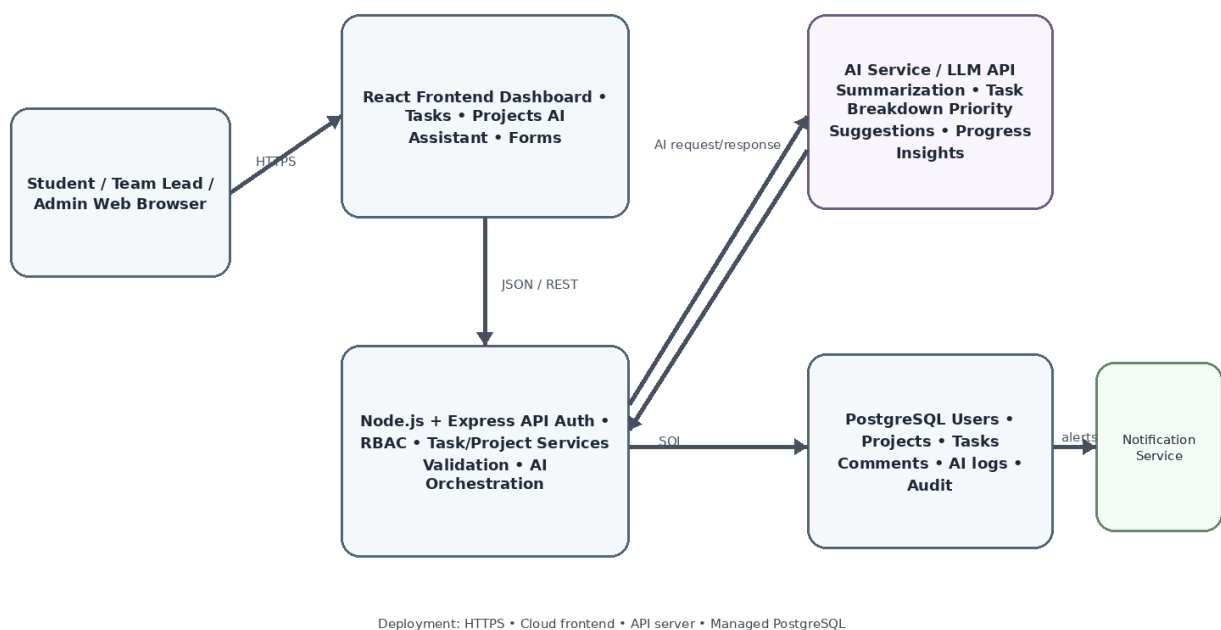
Week 1 Task: Project Planning and Architecture Design

AI-Assisted InternTrack — Intelligent Internship & Project Management System

Prepared by: Sanjeev Kumar

Role: Software Engineering Trainee — proposed focus: Backend, API & AI Integration
Internship Period: 14-Sep-2026 to 19-Oct-2026

AI-Assisted InternTrack — System Architecture



Week 1 Deliverable: Project Planning + System Architecture + AI Integration Plan

1. Project Overview

Because the team has not been assigned a project, this document proposes a realistic software-engineering project: an AI-Assisted Internship & Project Management System called InternTrack. It is a web application for students/trainees, Team Leads and Admins to manage projects, tasks, deadlines, progress and comments in one place, with AI used as an assistant for task breakdown, summaries and priority/risk suggestions.

1.1 Real-World Problem

Internship/project work can become difficult to track when assignments, deadlines, progress updates and blockers are scattered across chat messages, spreadsheets and documents. The proposed system creates a single source of truth for project work. The AI layer adds assistance without replacing the Team Lead or developer's decision-making.

1.2 Why This Is a Software Engineering Project

- Includes requirements engineering, scope control and stakeholder analysis.
- Uses layered architecture, REST APIs, database design and role-based access control.
- Requires SDLC planning, testing, version control, security and deployment.
- Contains defined modules and interfaces that can be developed and tested independently.
- AI is integrated as a software component with input validation, output validation, security and human approval.

2. Scope

In Scope

- Authentication and role-based access
- Project and milestone management
- Task creation, assignment and status workflow
- Comments/progress/blocker updates
- Dashboard, search and filters
- AI task breakdown
- AI project/task summaries
- AI priority/risk suggestions
- In-app notifications
- Audit history
- Responsive web interface

Out of Scope for MVP

- Autonomous AI decisions that change project status without approval
- Payroll or stipend management
- Video conferencing/chat replacement
- Native Android/iOS application
- Advanced predictive project forecasting
- Large enterprise multi-organization deployment

3. Stakeholders and Users

Stakeholder	Responsibility
Student / Trainee	View assigned tasks, update status, add comments and use AI assistance.
Team Lead	Create projects/tasks, assign work, review AI suggestions and approve project decisions.
Admin	Manage users, roles and system configuration.
Mentor / Sponsor	Review milestone and high-level progress.

4. Requirements Analysis

4.1 Functional Requirements

ID	Feature	Requirement
FR-01	Authentication	Secure login/logout and user account management.
FR-02	Role-based access	Student, Team Lead and Admin permissions.
FR-03	Project management	Create, update and track projects and milestones.
FR-04	Task management	Create, assign, edit, prioritize and close tasks.
FR-05	Task workflow	Planned → In Progress → Review → Completed with server-side validation.
FR-06	Comments	Add task comments and progress/blocker updates.
FR-07	Dashboard	Show progress, overdue work, workload and milestone status.
FR-08	Search/filter	Filter by project, assignee, status, priority and due date.
FR-09	AI task breakdown	Convert a task description into suggested smaller subtasks.
FR-10	AI summary	Generate a concise project/task progress summary from approved project data.
FR-11	AI priority suggestion	Suggest priority/risk level using task text, due date and project context; user confirms before saving.
FR-12	Notifications	In-app reminders for assignments, approaching deadlines and review requests.
FR-13	Audit log	Record important changes for traceability.

4.2 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	HTTPS, secure password hashing, server-side authorization and input validation.
NFR-02	AI safety	Do not send unnecessary sensitive data to an AI provider; show AI output as suggestions and require user confirmation for important changes.
NFR-03	Performance	Normal dashboard/API operations should target about 2 seconds under planned small-team load; AI responses may take longer and show loading state.
NFR-04	Usability	Responsive UI, clear status labels, simple navigation and readable AI suggestions.
NFR-05	Maintainability	Layered backend, reusable frontend components, API documentation and clean module boundaries.
NFR-06	Reliability	Validation, transactions and error handling should prevent inconsistent records.
NFR-07	Scalability	Architecture should allow additional projects/users without redesigning the entire application.
NFR-08	Compatibility	Current Chrome/Edge/Firefox on desktop and mobile-sized screens.

5. AI Use in the Project

AI will be implemented as an assistive service behind the backend API. The frontend will never call the AI provider directly. The backend selects the minimum necessary project/task context, sends a structured request, validates the returned format, and shows the result as a suggestion. Important actions such as changing priority or creating subtasks require user confirmation.

AI Feature	Input	Output	Benefit
AI Task Breakdown	Task title + description + constraints	Suggested subtasks	Turns large work into manageable units.
AI Progress Summary	Approved statuses + comments	Short summary + blockers	Reduces manual status-reporting effort.
AI Priority/Risk	Due date + task context + status	Priority/risk + reasons	Highlights work needing attention; user decides.
AI Project Assistant (future)	Approved project context	Answers status questions	Faster project information access.

5.1 AI Safety and Control

- AI output is advisory, not an automatic authority.
- Do not send passwords, authentication tokens or unnecessary personal data to the AI provider.
- Use server-side AI calls so provider credentials are not exposed in the browser.
- Validate structured AI output before displaying or saving it.
- Log AI-assisted actions where appropriate.
- Provide a non-AI fallback for core functions.

6. Proposed Technology Stack

Layer	Technology	Why
Frontend	React + TypeScript + CSS	Reusable components, forms, dashboard and AI assistant UI.

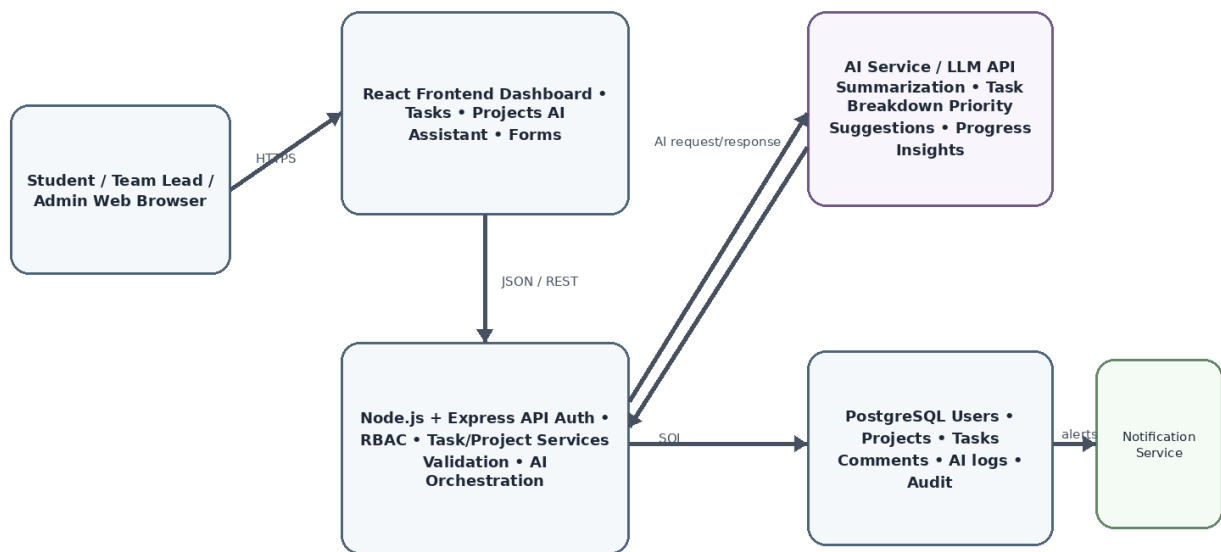
Layer	Technology	Why
Backend	Node.js + Express	REST APIs, business rules, validation and authorization.
Database	PostgreSQL	Relational storage, constraints, transactions and reporting queries.
AI Integration	LLM API through backend	Task breakdown, summaries and suggestions; backend controls prompts and data exposure.
Authentication	Secure session or JWT approach	Identity and role-based access.
Version Control	Git + GitHub	Branches, history, collaboration and review.
Testing	Jest/Supertest + UI testing	Unit, API, integration and functional checks.
Deployment	Cloud hosting + managed PostgreSQL	Staging/production deployment over HTTPS.

6.1 Why These Technologies?

- React provides reusable UI components for dashboards, forms, task cards and the AI assistant interface.
- Layered architecture separates presentation, business logic and data access, improving organization and testability.
- Node.js + Express provides a practical REST API layer for the MVP.
- PostgreSQL fits structured relational data such as users, projects, tasks, comments and audit records.
- Git/GitHub supports collaboration, branching, review and traceable changes.
- AI integration through the backend keeps provider keys and data-control logic away from the browser.

7. System Architecture

AI-Assisted InternTrack — System Architecture

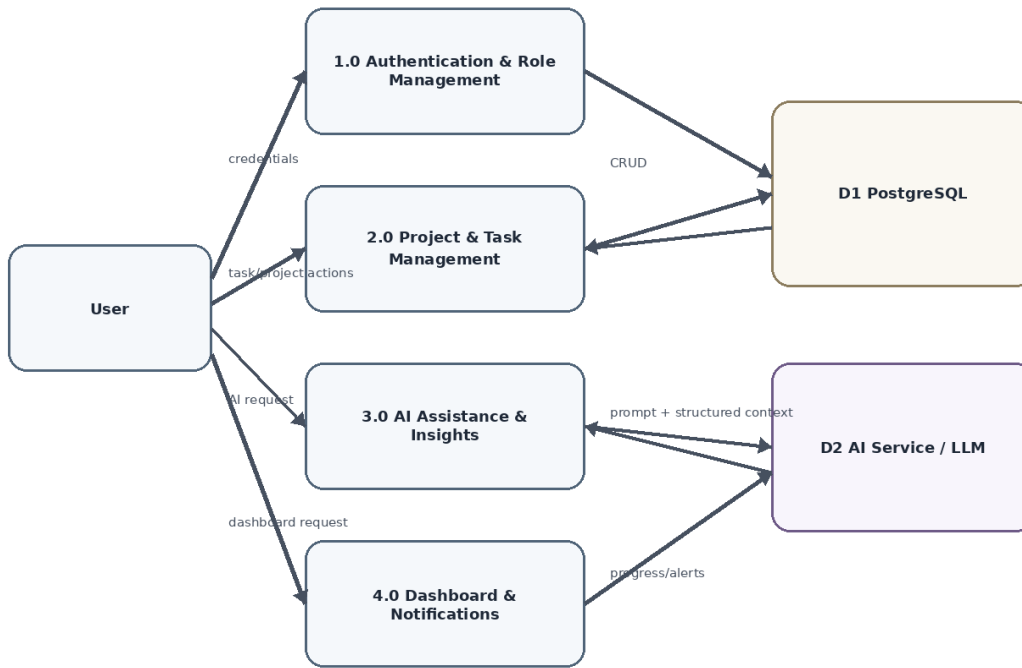


Deployment: HTTPS • Cloud frontend • API server • Managed PostgreSQL

Architecture flow: User → React Frontend → Node.js/Express REST API → PostgreSQL. The backend also communicates with the AI service for approved AI features. Notifications are generated by the backend. The MVP uses a modular monolith rather than microservices because the project is small and the team needs a simple architecture that can be developed and tested within five weeks.

8. Data Flow Diagram

AI-Assisted InternTrack — Level-1 Data Flow Diagram



Example AI flow: User requests task breakdown → backend authenticates/authorizes → backend selects approved task context → AI service returns structured suggestions → backend validates → frontend displays → user approves/edits → approved subtasks are saved.

9. Module Design

Module	Main Responsibilities
Authentication & RBAC	Login, roles and endpoint authorization
Project Module	Projects, milestones, members and status
Task Module	CRUD, assignment, priority, due date and workflow
AI Assistant Module	Task breakdown, summaries, priority/risk suggestions
Dashboard Module	Progress KPIs, overdue items and workload
Comment Module	Progress updates and blockers
Notification Module	Assignment/deadline/review reminders
Audit Module	Important changes and AI-assisted action traceability

10. Core Data Model

Entity	Important Fields	Relationship
User	id, name, email, password_hash, role, status	User can own/receive tasks and comments.
Project	id, name, description, start_date, end_date, status	One project contains many tasks/milestones.
Task	id, project_id, assignee_id, title, priority, status, due_date	Belongs to project; assigned to user.
Comment	id, task_id, user_id, body, created_at	Belongs to task; authored by user.
Notification	id, user_id, type, message, read_at	Belongs to user.
AIRequestLog	id, user_id, feature, created_at, status	Records AI feature usage without unnecessary sensitive prompts.
AuditLog	id, user_id, entity_type, entity_id, action, created_at	Tracks important changes.

11. Example REST API Contract

Method	Endpoint	Purpose
POST	/api/auth/login	Login
GET	/api/projects	List visible projects
POST	/api/projects	Create project

Method	Endpoint	Purpose
GET	/api/projects/:id/tasks	List project tasks
POST	/api/tasks	Create task
PATCH	/api/tasks/:id	Update task/status
POST	/api/tasks/:id/comments	Add comment
POST	/api/ai/task-breakdown	Generate suggested subtasks
POST	/api/ai/summary	Generate progress summary
POST	/api/ai/priority-suggestion	Suggest priority/risk
GET	/api/dashboard	Dashboard data

12. Roles and Permissions

Feature	Student	Team Lead	Admin
View own tasks	Yes	Yes	Yes
Update own task status	Yes	Yes	Yes
Create project	No	Yes	Yes
Create/assign tasks	No	Yes	Yes
Use AI task breakdown	Yes	Yes	Yes
Approve AI priority change	No	Yes	Yes
View team progress	Limited	Yes	Yes
Manage users/roles	No	Limited	Yes

13. Project Roadmap and Timeline

Week	Dates	Milestone	Main Work	Deliverable
Week 1	14–20 Sep	Planning & Architecture	Research, problem definition, scope, requirements, AI use cases, architecture, DFD, risks	PDF/DOCX planning package
Week 2	21–27 Sep	Foundation	Repo, database schema, backend structure, authentication/RBAC, API contract	Working foundation
Week 3	28 Sep–4 Oct	Core Features	Projects, tasks, assignments, status workflow, comments	Core task management
Week 4	5–11 Oct	AI + Dashboard + Testing	AI service integration, dashboard, filters, notifications, API/UI tests	Integrated MVP
Week 5	12–19 Oct	Deployment & Finalization	Security review, deployment, bug fixes, documentation and demo	Final build + documentation

14. Week 1 Work Breakdown — 32 Hours

Activity	Hours
Problem research and project selection	4
Scope + stakeholders	3
Functional requirements	4
Non-functional + AI requirements	3
Technology/architecture design	4
Architecture + DFD diagrams	5
Data model + API planning	3
Timeline + assumptions + risks	2
Final review/formatting	4
Total	32

15. Assumptions

- Team has not been assigned a project, so InternTrack is a proposed project for Week 1 and must be approved by the Team Lead.
- The MVP is intended for a small internship/project team, not a public enterprise deployment.
- AI features provide assistance, not autonomous decision-making.

- Only minimum necessary project data is sent to the AI service.
- Email notification is optional; in-app notifications are part of the MVP.
- No proprietary dataset is required; the application works mainly with user-entered project/task data.

16. Constraints

- Week 1 assignment expects approximately 30–35 hours.
- Five-week internship window limits the MVP scope.
- Small trainee team and limited budget.
- AI provider availability, latency and usage limits may affect AI features.
- Security/privacy requirements restrict data sent to external AI services.

17. Risks and Mitigation

ID	Risk	Impact	Mitigation
R1	AI output may be inaccurate	High	Treat AI output as a suggestion; user confirms; validate structured output.
R2	Sensitive data sent to AI	High	Minimize data, redact unnecessary fields, keep AI calls server-side, document provider/data policy.
R3	Scope becomes too large	High	Freeze MVP: task breakdown + summaries first; advanced AI features remain optional.
R4	Security/authorization bug	High	Enforce authorization at every protected API endpoint and test negative cases.
R5	API integration failure	Medium	Define API schema early; mock AI service during development/testing.
R6	Timeline delay	Medium	Weekly milestones, daily blocker updates and small deliverables.
R7	AI cost/rate limits	Medium	Limit calls, cache summaries where appropriate, provide non-AI fallback.
R8	Database changes	Medium	Design schema early and use versioned migrations.

18. Testing Strategy

- Unit testing: task rules, priority rules, AI response parsing and utility functions.
- API testing: authentication, authorization, CRUD, validation, error handling and AI endpoints.
- Integration testing: frontend → API → database and backend → AI service.
- AI testing: normal, empty, malformed, adversarial and oversized inputs; validate structured output.
- UI testing: login, dashboard, task creation, status, AI suggestions, approval/edit flow and notifications.
- Security testing: unauthorized access, role escalation, input validation and token/session handling.
- Regression testing before each milestone.

19. Security Design

Protected API endpoints must check authorization on the server. Secure REST services should use HTTPS, and access control should be enforced at each endpoint. Workflow state should be validated on the server rather than relying only on frontend logic. These principles are especially important because the application contains role-based actions and AI-assisted workflows.

20. Future Enhancements

- AI-generated sprint planning suggestions
- Calendar/Gantt view
- File attachments
- Team chat integration
- Advanced workload analytics
- AI-assisted meeting/action-item summaries
- Mobile app
- Multi-organization support

21. Expected Benefits

- Centralized project/task management
- Clear task ownership and deadlines
- Faster progress reporting
- AI-assisted breakdown of complex tasks
- Early visibility of possible blockers
- Practical demonstration of the full software engineering lifecycle
- Separation between core deterministic functions and optional AI assistance

22. Conclusion

InternTrack is a feasible five-week software engineering project for a trainee team that currently has no assigned project. It covers the complete planning foundation requested in Week 1: problem research, scope, requirements, design strategy, architecture, data flow, timeline, assumptions, constraints and risks. AI is included as a controlled assistant rather than the core dependency, so the system remains useful even when AI is unavailable.

23. Public Resources Used for Guidance

- React documentation — reusable components and UI architecture: <https://react.dev/learn>
- Microsoft Learn — layered architecture and separation of concerns:
<https://learn.microsoft.com/en-us/dotnet/architecture/modern-web-apps-azure/common-web-application-architectures>
- OWASP REST Security Cheat Sheet — HTTPS, endpoint authorization, workflow validation and audit logging:
https://cheatsheetseries.owasp.org/cheatsheets/REST_Security_Cheat_Sheet.html
- OWASP Authorization Cheat Sheet — authentication vs authorization and access control:
https://cheatsheetseries.owasp.org/cheatsheets/Authorization_Cheat_Sheet.html
- Microsoft Learn — application architecture design concepts:
<https://learn.microsoft.com/en-us/training/modules/design-application-architecture/>

Appendix A — What to Tell the Team Lead

- “Sir, since our team has not been assigned a project, I have prepared a project proposal called AI-Assisted InternTrack.”
- “It is a web application for managing internship/project tasks, deadlines, progress and team communication in one place.”
- “AI will be an assistant: it can break a task into subtasks, summarize project progress and suggest priority/risk. Final decisions remain with the user or Team Lead.”
- “Our main stack is React + TypeScript on the frontend, Node.js + Express for REST APIs, PostgreSQL for the database and an AI API integrated through the backend.”
- “The architecture is Frontend → Backend API → Database, with the backend also controlling AI requests. This keeps the project modular and easier to test.”
- “My proposed role is Software Engineering Trainee with a Backend/API & AI Integration focus. I can work on API design, database integration, AI service integration, validation, testing and documentation; exact ownership can be finalized by the Team Lead.”
- “Week 1 is planning and architecture. Week 2 authentication/database foundation, Week 3 core task management, Week 4 AI/dashboard/testing and Week 5 deployment/documentation.”
- “We will keep the MVP limited so the team can realistically complete it in the internship period.”

Appendix B — Your Work Role in Detail

Area	Your Responsibility
Requirements	Understand requirements, clarify acceptance criteria and document them.
Backend/API	Design REST endpoints, request/response structure, validation and business rules.
Database	Help design PostgreSQL tables, relationships, indexes and migrations.
AI Integration	Build backend service for AI calls, prompt/context control, output validation and error handling.

Area	Your Responsibility
Security	Implement/verify role checks, secure authentication flow and protected endpoints.
Testing	Write API/unit tests and test AI edge cases and failure responses.
Documentation	Maintain API notes, architecture decisions and final technical documentation.
Team Collaboration	Use Git branches, pull requests, issue tracking and progress updates.

Appendix C — Questions for Team Lead

- Is this proposed project approved for our Week 1 submission?
- Which exact module should I own after the team divides work?
- Do we have a preferred frontend/backend/database stack?
- Which AI provider/API is allowed by the organization?
- Are there restrictions on sending project/user data to an external AI provider?
- What Git branching and pull-request process should we follow?
- What should be the acceptance criteria for Week 2?